

Java OOP Advanced Activity: PayWave Payment System**1. Code Screenshots****PaymentMethod.java**

```
JAVA > PaywavePaymentSystem > PaymentMethod.java > PaymentMethod
1  package PaywavePaymentSystem;
2
3  public interface PaymentMethod {
4
5      /*
6       * DOCU: Processes a payment of the specified amount
7       * @param {double} amount - The payment amount to be processed
8       * @returns {boolean} - True if the payment was successful, false otherwise
9       *
10     * Last Updated: 2026-03-09
11     * Author: Louis Zane De Jesus
12     * Last Updated By: Louis Zane De Jesus
13     */
14     boolean processPayment(double amount);
15
16     /*
17     * DOCU: Retrieves the payment details of the payment method
18     * @returns {String} - A formatted string containing the payment method details
19     *
20     * Last Updated: 2026-03-09
21     * Author: Louis Zane De Jesus
22     * Last Updated By: Louis Zane De Jesus
23     */
24     String getPaymentDetails();
25 }
```

Refundable.java

```
JAVA > PaywavePaymentSystem > Refundable.java > Refundable > processRefund(double)
1  package PaywavePaymentSystem;
2
3  public interface Refundable {
4
5      /*
6       * DOCU: Processes a refund of the specified amount back to the payment method
7       * @param {double} amount - The refund amount to be returned
8       * @returns {boolean} - True if the refund was successful, false otherwise
9       *
10     * Last Updated: 2026-03-09
11     * Author: Louis Zane De Jesus
12     * Last Updated By: Louis Zane De Jesus
13     */
14     boolean processRefund(double amount);
15
16     /*
17     * DOCU: Retrieves the payment details of the refundable payment method
18     * @returns {String} - A formatted string containing the payment method details
19     *
20     * Last Updated: 2026-03-09
21     * Author: Louis Zane De Jesus
22     * Last Updated By: Louis Zane De Jesus
23     */
24     String getPaymentDetails();
25 }
```

EWalletPayment.java

```
JAVA > PaywavePaymentSystem > EWalletPayment.java > EWalletPayment > EWalletPayment(String, double)
1 package PaywavePaymentSystem;
2
3 public class EWalletPayment implements PaymentMethod {
4
5     private String walletID;
6     private double balance;
7
8     /*
9      * DOCU: Constructs an EWalletPayment with the given wallet ID and initial balance
10     * @param {String} walletID - The unique identifier for the e-wallet
11     * @param {double} balance - The initial balance of the e-wallet
12     *
13     * Last Updated: 2026-03-09
14     * Author: Louis Zane De Jesus
15     * Last Updated By: Louis Zane De Jesus
16     */
17     public EWalletPayment(String walletID, double balance) {
18         this.walletID = walletID;
19         this.balance = balance;
20     }
21
22     /*
23     * DOCU: Processes an e-wallet payment by deducting the amount from the balance
24     * @param {double} amount - The payment amount to be deducted
25     * @returns {boolean} - True if payment succeeded, false if amount is invalid or exceeds balance
26     *
27     * Last Updated: 2026-03-09
28     * Author: Louis Zane De Jesus
29     * Last Updated By: Louis Zane De Jesus
30     */
31     @Override
32     public boolean processPayment(double amount) {
33         if (amount <= 0 || amount > balance) {
34             System.out.println("Payment failed: Insufficient E-Wallet balance.");
35             return false;
36         }
37         balance -= amount;
38         System.out.println("Payment processed successfully.");
39         return true;
40     }
41 }
```

```
42 /*
43  * DOCU: Returns the e-wallet payment details as a formatted string
44  * @returns {String} - A string containing the wallet ID and current balance
45  *
46  * Last Updated: 2026-03-09
47  * Author: Louis Zane De Jesus
48  * Last Updated By: Louis Zane De Jesus
49  */
50 @Override
51 public String getPaymentDetails() {
52     return "EWallet [walletID=" + walletID + ", balance=" + balance + "];"
53 }
54 }
55 }
```

CreditCardPayment.java

```
JAVA > PaywavePaymentSystem > CreditCardPayment.java > CreditCardPayment > CreditCardPayment(String, double)
1  package PaywavePaymentSystem;
2
3  public class CreditCardPayment implements PaymentMethod, Refundable {
4
5      private String cardNumber;
6      private double creditLimit;
7      private double availableCredit;
8
9      /**
10     * DOCU: Constructs a CreditCardPayment with the given card number and credit limit
11     * @param {String} cardNumber - The credit card number
12     * @param {double} creditLimit - The maximum credit limit for this card
13     *
14     * Last Updated: 2026-03-09
15     * Author: Louis Zane De Jesus
16     * Last Updated By: Louis Zane De Jesus
17     */
18     public CreditCardPayment(String cardNumber, double creditLimit) {
19         this.cardNumber = cardNumber;
20         this.creditLimit = creditLimit;
21         this.availableCredit = creditLimit;
22     }
23
24     /**
25     * DOCU: Processes a credit card payment by deducting the amount from available credit
26     * @param {double} amount - The payment amount to be charged
27     * @returns {boolean} - True if payment succeeded, false if amount is invalid or exceeds available credit
28     *
29     * Last Updated: 2026-03-09
30     * Author: Louis Zane De Jesus
31     * Last Updated By: Louis Zane De Jesus
32     */
33     @Override
34     public boolean processPayment(double amount) {
35         if (amount <= 0 || amount > availableCredit) {
36             System.out.println("Payment failed: Insufficient Credit Card balance.");
37             return false;
38         }
39         availableCredit -= amount;
40         System.out.println("Payment processed successfully.");
41         return true;
42     }
43
44     /**
45     * DOCU: Processes a refund by adding the amount back to the available credit
46     * @param {double} amount - The refund amount to be credited back
47     * @returns {boolean} - True if the refund was successful
48     *
49     * Last Updated: 2026-03-09
50     * Author: Louis Zane De Jesus
51     * Last Updated By: Louis Zane De Jesus
52     */
53     @Override
54     public boolean processRefund(double amount) {
55         availableCredit += amount;
56         return true;
57     }
58
59     /**
60     * DOCU: Returns the credit card payment details as a formatted string
61     * @returns {String} - A string containing the card number and available credit
62     *
63     * Last Updated: 2026-03-09
64     * Author: Louis Zane De Jesus
65     * Last Updated By: Louis Zane De Jesus
66     */
67     @Override
68     public String getPaymentDetails() {
69         return "CreditCard [cardNumber=" + cardNumber + ", availableCredit=" + availableCredit + "];";
70     }
71 }
```

BankTransferPayment.java

```
JAVA > PaywavePaymentSystem > BankTransferPayment.java > BankTransferPayment > processPayment(double)
1 package PaywavePaymentSystem;
2
3 public class BankTransferPayment implements PaymentMethod, Refundable {
4
5     private String accountNumber;
6     private double balance;
7
8     /*
9     * DOCU: Constructs a BankTransferPayment with the given account number and balance
10    * @param {String} accountNumber - The bank account number
11    * @param {double} balance - The initial balance of the bank account
12    *
13    * Last Updated: 2026-03-09
14    * Author: Louis Zane De Jesus
15    * Last Updated By: Louis Zane De Jesus
16    */
17    public BankTransferPayment(String accountNumber, double balance) {
18        this.accountNumber = accountNumber;
19        this.balance = balance;
20    }
21
22    /*
23    * DOCU: Processes a bank transfer payment by deducting the amount from the balance
24    * @param {double} amount - The payment amount to be transferred
25    * @returns {boolean} - True if payment succeeded, false if amount is invalid or exceeds balance
26    *
27    * Last Updated: 2026-03-09
28    * Author: Louis Zane De Jesus
29    * Last Updated By: Louis Zane De Jesus
30    */
31    @Override
32    public boolean processPayment(double amount) {
33        if (amount <= 0 || amount > balance) {
34            System.out.println(x: "Payment failed: Insufficient Bank Transfer balance.");
35            return false;
36        }
37        balance -= amount;
38        System.out.println(x: "Payment processed successfully.");
39        return true;
40    }
41
42    /*
43    * DOCU: Processes a refund by adding the amount back to the bank account balance
44    * @param {double} amount - The refund amount to be credited back
45    * @returns {boolean} - True if the refund was successful
46    *
47    * Last Updated: 2026-03-09
48    * Author: Louis Zane De Jesus
49    * Last Updated By: Louis Zane De Jesus
50    */
51    @Override
52    public boolean processRefund(double amount) {
53        balance += amount;
54        System.out.println(x: "Refund processed successfully.");
55        return true;
56    }
57
58    /*
59    * DOCU: Returns the bank transfer payment details as a formatted string
60    * @returns {String} - A string containing the account number and current balance
61    *
62    * Last Updated: 2026-03-09
63    * Author: Louis Zane De Jesus
64    * Last Updated By: Louis Zane De Jesus
65    */
66    @Override
67    public String getPaymentDetails() {
68        return "BankTransfer [accountNumber=" + accountNumber + ", balance=" + balance + "];"
69    }
70 }
```

Sample Output

```
Payment processed successfully.  
CreditCard [cardNumber=CC1234, availableCredit=0.0]  
Payment failed: Insufficient E-Wallet balance.  
EWallet [walletID=EW9876, balance=200.0]  
Payment processed successfully.  
BankTransfer [accountNumber=BA1122, balance=7000.0]  
PS C:\Users\louis\Git\My_First_Repo> 
```